

ER Modeling II

ER Structure

English grammar structure ER structure

Common noun Entity type

Proper noun Entity

Transitive verb Relationship type

Intransitive verb Attribute type

Adjective Attribute for entity

Adverb Attribute for relationship

Associative Entity

- An **associative entity** is an element of the [entity-relationship model](#). The database [relational model](#) does not offer direct support to [many-to-many relationships](#), even though such relationships happen frequently in normal usage. The solution to this problem is the creation of another [table](#) to hold the necessary information for this relationship. This new table is called an **associative entity**.
- An associative entity can be thought of as both an entity and a relationship since it encapsulates properties from both. It is a relationship since it is serving to join two or more entities together, but it is also an entity since it may have its own properties. The associative entity must have the primary keys of both adjoining tables as identifiers, but may also contain its own unique identifier and other information about the relationship

Associative Entity



Example I

- A college contains many departments
- Each department can offer any number of courses
- Many instructors can work in a department
- An instructor can work only in one department
- For each department there is a Head
- An instructor can be head of only one department
- Each instructor can take any number of courses
- A course can be taken by only one instructor
- A student can enroll for any number of courses
- Each course can have any number of students

Example I

Identify the Entities

What are the entities here?

From the statements given, the entities are

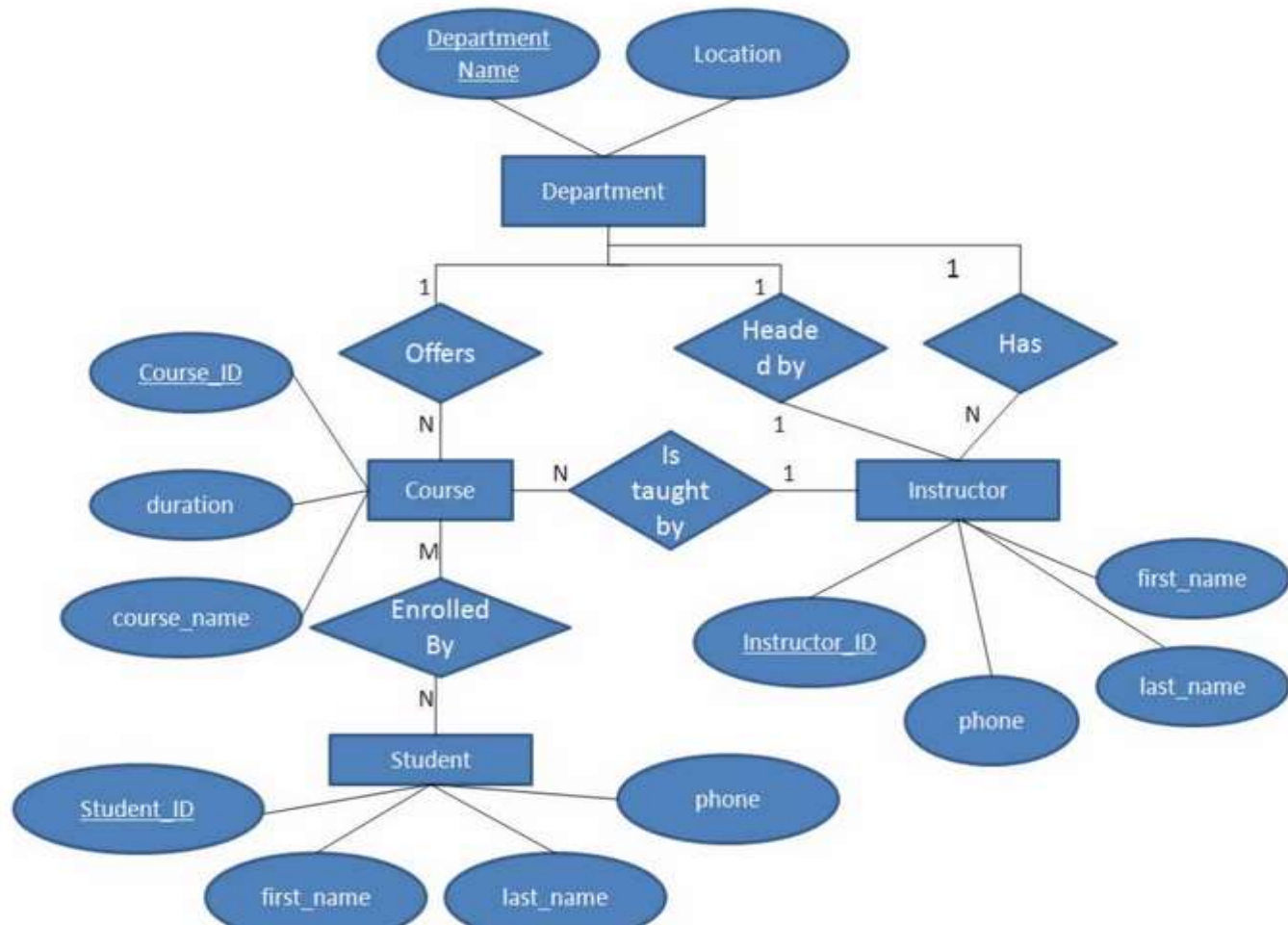
- Department
- Course
- Instructor
- Student

Example I

Identify the relationships

- One department offers many courses. But one particular course can be offered by only one department. hence the cardinality between department and course is One to Many (1:N)
- One department has multiple instructors . But instructor belongs to only one department. Hence the cardinality between department and instructor is One to Many (1:N)
- One department has only one head and one head can be the head of only one department. Hence the cardinality is one to one. (1:1)
- One course can be enrolled by many students and one student can enroll for many courses. Hence the cardinality between course and student is Many to Many (M:N)
- One course is taught by only one instructor. But one instructor teaches many courses. Hence the cardinality between course and instructor is Many to One (N :1)

ER-Model-I



Example II

- A company database needs to store information about employees (identified by *ssn*, with *salary* and *phone*) departments (identified by *dno*, with *dname* and *budget*), and children of employees (with *name* and *age*).
- Employees *work in departments*; each department is managed by an employee; a child must be identified uniquely by *name* when the parent (*who is an employee*; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company.

Example II

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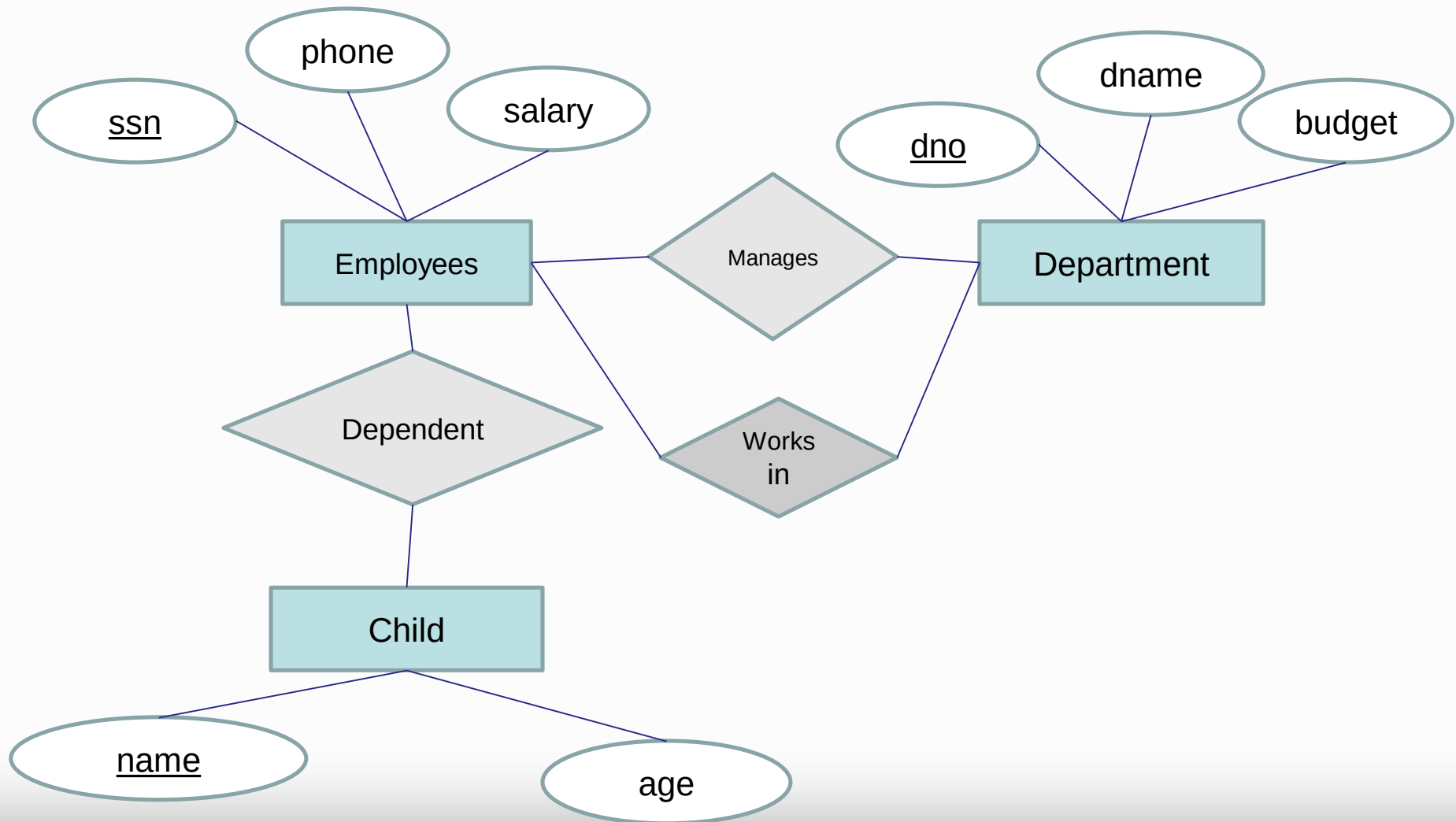
Example II

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ER-Model-II



Example III

- Galleries keep information about artists, their names (which are unique), birthplaces, age, and style of art. For each piece of artwork, the artist, the year it was made, its unique title, its type of art (e.g., painting, sculpture, photograph), and its price must be stored. Pieces of artwork are also classified into groups of various kinds, for example, portraits works by Picasso, or works of the 19th century; a given piece may belong to more than one group.
- Each group is identified by a name (like those just given) that describes the group. Finally, galleries keep information about customers. For each customer, galleries keep that person's unique name, address, total amount of dollars spent in the gallery (very important!), and the artists and groups of art that the customer tends to like.
- Draw the ER diagram for the database.

Example III- Entites

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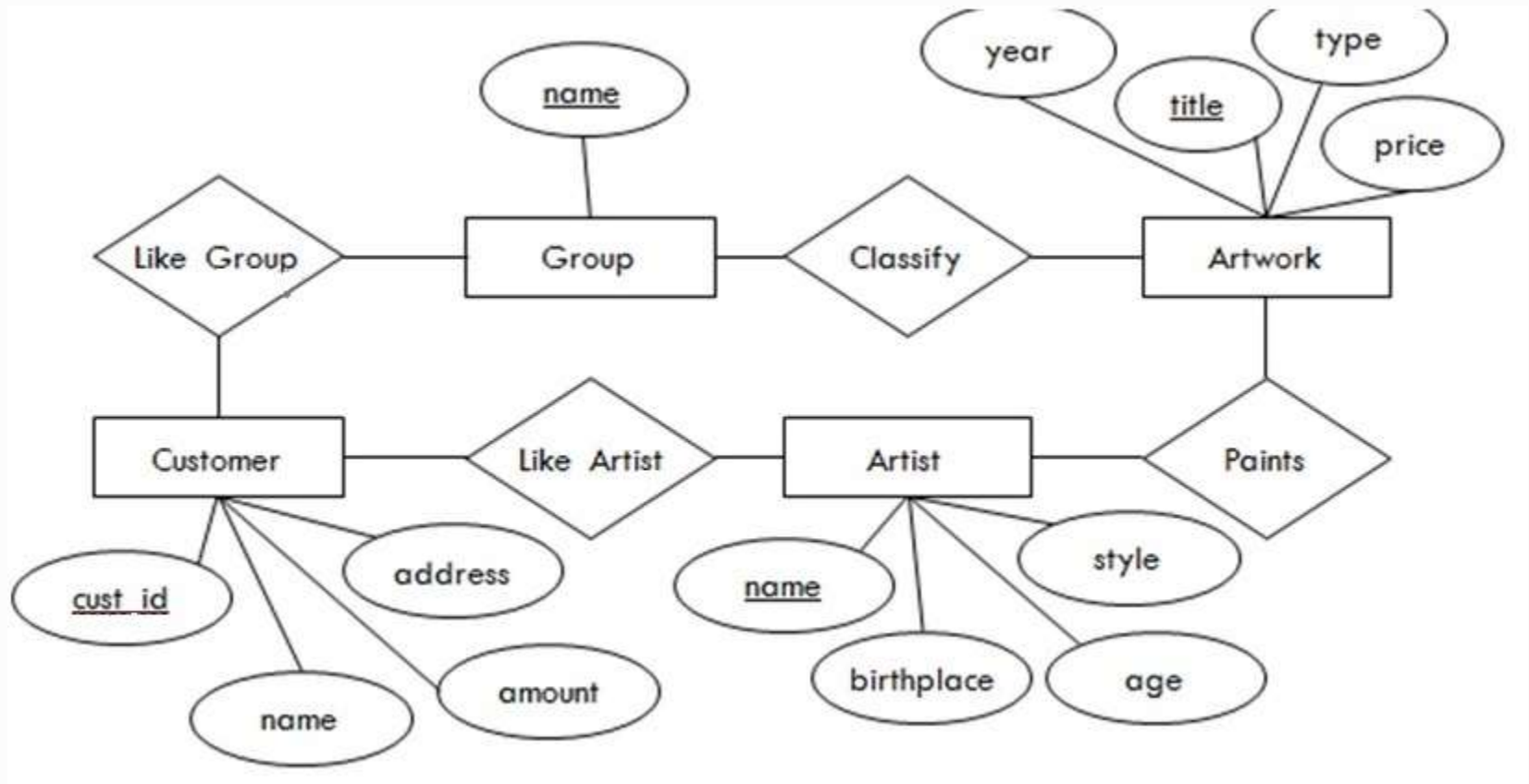
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ER-Model-III



Case Study I-*DIY*

In our database, we have students. They have a name, a registration number, and a course of study. The university offers lectures. Each lecture may be part of some course of study in a certain semester. Lectures may have other lectures as prerequisites. They have a title, provide a specific number of credits and have a unique ID. Each year, some of these lectures are offered by a professor at a certain day at a fixed time in a specific room. Students may register for that lecture. Professors have a name and are member of a specific department.

Case Study II-*DIY*

The craft trading website ABC is setting up a database to record sellers and their products.

This requires recording the following information:

- For each seller, their name, contact email, and postal address.
- For each product, its name, price, and number available.
- Which product is from which seller.
- A unique id number for each product.

Draw an entity-relationship (ER) diagram that represents this information.

Make sure to capture the constraints on the relationships involved, and designate appropriate primary keys for the entities.